2.06 Sequences and Series: Introduction

Worked Solutions

1(a))

Divergent as the distances between the terms are not decreasing.

1(b))

Periodic as the numbers 1 and 5 are repeating. This sequence has period 2.

1(c))

Convergent as the distances between the terms is decreasing.

1(d))

Convergent as the distances between the terms is decreasing.

2(a))

$$u_1 = 5 \times 1 - 1 = 4 \qquad u_2 = 5 \times 2 - 1 = 9$$
$$u_3 = 5 \times 3 - 1 = 14 \qquad u_4 = 5 \times 4 - 1 = 19$$
$$\Sigma = 4 + 9 + 14 + 19 = 46$$

2(b))

$$u_1 = 3^1 + 1^2 = 4 \qquad u_2 = 3^2 + 2^2 = 13 \qquad u_3 = 3^3 + 3^2 = 36$$
$$\Sigma = 4 + 13 + 36 = 53$$

2(c))

$$u_1 = \frac{2}{1} = 2 \qquad u_2 = \frac{2}{2} = 1 \qquad u_3 = \frac{2}{3} \qquad u_4 = \frac{2}{4} = \frac{1}{2} \qquad u_5 = \frac{2}{5}$$
$$\Sigma = 2 + 1 + \frac{2}{3} + \frac{1}{2} + \frac{2}{5} = \frac{137}{30}$$

2(d))

$$u_1 = 4 \times \left(\frac{1}{2}\right)^{1-1} = 4 \qquad u_2 = 4 \times \left(\frac{1}{2}\right)^{2-1} = 2$$
$$u_3 = 4 \times \left(\frac{1}{2}\right)^{3-1} = 1 \qquad u_4 = 4 \times \left(\frac{1}{2}\right)^{4-1} = \frac{1}{2}$$
$$\Sigma = 4 + 2 + 1 + \frac{1}{2} = \frac{15}{2}$$

3(a))

$$u_3 = 3^2 - 10 = -1 \qquad u_4 = 4^2 - 10 = 6$$
$$u_5 = 5^2 - 10 = 15 \qquad u_6 = 6^2 - 10 = 26$$
$$\Sigma = (-1) + 6 + 15 + 26 = 46$$

3(b))

$$u_{99} = \frac{3}{99} = \frac{1}{33} \qquad u_{100} = \frac{3}{100}$$
$$\Sigma = \frac{1}{33} + \frac{3}{100} = \frac{199}{3300}$$

3(c))

$$u_6 = 50 \times 5^{6-5} = 250 \qquad u_7 = 50 \times 5^{7-5} = 1250 \qquad u_8 = 50 \times 5^{8-5} = 6250$$
$$\Sigma = 250 + 1250 + 6250 = 7750$$

4(a))

$$u_1 = 1 + k \qquad u_2 = 2 + k \qquad u_3 = 3 + k \qquad u_4 = 4 + k \qquad u_5 = 5 + k$$
$$\Sigma = (1 + k) + (2 + k) + (3 + k) + (4 + k) + (5 + k) = 15 + 5k$$

4(b))

$$u_1 = 1 - k \times 1^3 = 1 - k \qquad u_2 = 1 - k \times 2^3 = 1 - 8k$$
$$u_3 = 1 - k \times 3^3 = 1 - 27k \qquad u_4 = 1 - k \times 4^3 = 1 - 64k$$
$$\Sigma = (1 - k) + (1 - 8k) + (1 - 27k) + (1 - 64k) = 4 - 100k$$

4(c))

$$u_2 = \frac{k}{2+1} = \frac{1}{3}k \qquad u_3 = \frac{k}{3+1} = \frac{1}{4}k \qquad u_4 = \frac{k}{4+1} = \frac{1}{5}k$$
$$\Sigma = \frac{1}{3}k + \frac{1}{4}k + \frac{1}{5}k = \frac{47}{60}k$$

4(d))

$$u_3 = \frac{k \times 3}{k+1} = \frac{3k}{k+1} \qquad u_4 = \frac{k \times 4}{k+1} = \frac{4k}{k+1}$$
$$u_5 = \frac{k \times 5}{k+1} = \frac{5k}{k+1} \qquad u_6 = \frac{k \times 6}{k+1} = \frac{6k}{k+1}$$
$$\Sigma = \frac{3k}{k+1} + \frac{4k}{k+1} + \frac{5k}{k+1} + \frac{6k}{k+1} = \frac{18k}{k+1}$$

5)

$$\sum_{n=20}^{24} f(n) = \sum_{n=1}^{24} f(n) - \sum_{n=1}^{19} f(n) = 3 - \frac{19}{10} = \frac{11}{10}$$

6(a))

$$u_1 = (1+2)^2 = 9 \qquad u_2 = (2+2)^2 = 16$$
$$g(2) = 9 + 16 = 25$$

6(b))

$$u_1 = (1+2)^2 = 9 \qquad u_2 = (2+2)^2 = 16 \qquad u_3 = (3+2)^2 = 25$$
$$u_4 = (4+2)^2 = 36 \qquad u_5 = (5+2)^2 = 49$$
$$g(5) = 9 + 16 + 25 + 36 + 49 = 135$$

6(c))

$$u_1 = (1+2)^2 = 9 \qquad u_2 = (2+2)^2 = 16 \qquad u_3 = (3+2)^2 = 25$$
$$u_4 = (4+2)^2 = 36 \qquad u_5 = (5+2)^2 = 49 \qquad u_6 = (6+2)^2 = 64 \qquad u_7 = (7+2)^2 = 81$$
$$g(7) = 9 + 16 + 25 + 36 + 49 + 64 + 81 = 280$$
$$g(6) = 9 + 16 + 25 + 36 + 49 + 64 = 199$$
$$\therefore g(7) - g(6) = 280 - 199 = 81$$

i.e. $\sum_{n=1}^{7}(n+2)^2 - \sum_{n=1}^{6}(n+2)^2 = u_7$.

6(d))

$$g(k+1) - g(k) = \sum_{n=1}^{k+1}(n+2)^2 - \sum_{n=1}^{k}(n+2)^2 = u_{k+1} = ((k+1)+2)^2 = (k+3)^2$$

7)

$$\sum_{i=1}^{n} 1 = n$$

$$\sum_{i=1}^{100} 1 + \sum_{i=20}^{100} 1 + \sum_{i=50}^{80} 1 = 100 + (100 - 19) + (80 - 49) = 212$$

8)

$$u_1 = \frac{6(1+1)}{\sqrt{1}} = 12$$
$$u_2 = \frac{6(2+1)}{\sqrt{2}} = \frac{18}{\sqrt{2}} = \frac{18\sqrt{2}}{2} = 9\sqrt{2}$$
$$u_3 = \frac{6(3+1)}{\sqrt{3}} = \frac{24}{\sqrt{3}} = \frac{24\sqrt{3}}{3} = 8\sqrt{3}$$
$$u_4 = \frac{6(4+1)}{\sqrt{4}} = \frac{30}{2} = 15$$
$$\Sigma = 12 + 9\sqrt{2} + 8\sqrt{3} + 15 = 27 + 9\sqrt{2} + 8\sqrt{3}$$

9)

$$u_1 = \frac{1}{\sqrt{1}+\sqrt{1+1}} = \frac{1}{1+\sqrt{2}} \times \frac{1-\sqrt{2}}{1-\sqrt{2}} = \frac{1-\sqrt{2}}{1-2} = \frac{1-\sqrt{2}}{-1} = -1+\sqrt{2}$$
$$u_2 = \frac{2}{\sqrt{2}+\sqrt{2+1}} = \frac{2}{\sqrt{2}+\sqrt{3}} \times \frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}-\sqrt{3}} = \frac{2\sqrt{2}-2\sqrt{3}}{2-3} = \frac{2\sqrt{2}-2\sqrt{3}}{-1} = -2\sqrt{2}+2\sqrt{3}$$
$$u_3 = \frac{3}{\sqrt{3}+\sqrt{3+1}} = \frac{3}{\sqrt{3}+2} \times \frac{\sqrt{3}-2}{\sqrt{3}-2} = \frac{3\sqrt{3}-6}{3-4} = \frac{3\sqrt{3}-6}{-1} = -3\sqrt{3}+6$$
$$\Sigma = (-1+\sqrt{2}) + (-2\sqrt{2}+2\sqrt{3}) + (-3\sqrt{3}+6) = 5 - \sqrt{2} - \sqrt{3}$$

10)

$$u_1 = \ln(1+1) = \ln 2 \qquad u_2 = \ln(2+1) = \ln 3 \qquad u_3 = \ln(3+1) = \ln 4$$
$$\Sigma = \ln 2 + \ln 3 + \ln 4 = \ln(2 \times 3 \times 4) = \ln 24$$

11)

$$u_1 = \ln\left(\frac{1}{1}\right) = \ln 1 = 0 \qquad u_2 = \ln\left(\frac{1}{2}\right) \qquad u_3 = \ln\left(\frac{1}{3}\right) \qquad u_4 = \ln\left(\frac{1}{4}\right)$$
$$\Sigma = 0 + \ln\left(\frac{1}{2}\right) + \ln\left(\frac{1}{3}\right) + \ln\left(\frac{1}{4}\right) = \ln\left(\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4}\right)$$
$$= \ln\left(\frac{1}{24}\right) = \ln(24^{-1}) = -1 \times \ln 24 = -\ln 24$$

12)

$$u_1 = \ln\left(\frac{1+1}{1}\right) = \ln 2 \qquad u_2 = \ln\left(\frac{2+1}{2}\right) = \ln\left(\frac{3}{2}\right)$$
$$u_3 = \ln\left(\frac{3+1}{3}\right) = \ln\left(\frac{4}{3}\right) \qquad u_4 = \ln\left(\frac{4+1}{4}\right) = \ln\left(\frac{5}{4}\right)$$
$$u_5 = \ln\left(\frac{5+1}{5}\right) = \ln\left(\frac{6}{5}\right)$$
$$\Sigma = \ln 2 + \ln\left(\frac{3}{2}\right) + \ln\left(\frac{4}{3}\right) + \ln\left(\frac{5}{4}\right) + \ln\left(\frac{6}{5}\right) = \ln\left(2 \times \frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \frac{6}{5}\right) = \ln 6$$

13)

$$h(k+1)-h(k)=\sum_{n=1}^{k+1}\frac{n}{n+1}-\sum_{n=1}^{k}\frac{n}{n+1}=u_{k+1}=\frac{k+1}{(k+1)+1}=\frac{k+1}{k+2}$$

$$\frac{k+1}{k+2}=\frac{99}{100}\Rightarrow k+1=\frac{99}{100}(k+2)$$

$$\Rightarrow 100(k+1)=99(k+2)\Rightarrow 100k+100=99k+198\Rightarrow k=98$$

Note: you may spot $k=98$ is the answer earlier!

14)

$$u_1=50$$

$$u_2=3u_1+4=3\times 50+4=154$$

$$u_3=3u_2+4=3\times 154+4=466$$

$$u_4=3u_3+4=3\times 466+4=1402$$

15)

$$u_1=1000$$

$$u_2=\frac{u_1+2}{10}=\frac{1000+2}{10}=\frac{501}{5}$$

$$u_3=\frac{u_2+2}{10}=\frac{\frac{501}{5}+2}{10}=\frac{511}{50}$$

$$u_4=\frac{u_3+2}{10}=\frac{\frac{511}{50}+2}{10}=\frac{611}{500}$$

16)

$$u_1=2\qquad u_2=5u_1+k=5\times 2+k=10+k$$

$$u_3=5u_2+k=5(10+k)+k=50+6k$$

$$\therefore 50+6k=4\Rightarrow k=-\frac{23}{3}$$

17)

$$u_1=3\qquad u_2=\frac{1}{u_1+2}=\frac{1}{3+2}=\frac{1}{5}\qquad u_3=\frac{1}{u_2+2}=\frac{1}{\frac{1}{5}+2}=\frac{5}{11}$$

$$\Sigma=3+\frac{1}{5}+\frac{5}{11}=\frac{201}{55}$$

18)

$$u_1 = 10 \qquad u_2 = \frac{10}{u_1} - k = \frac{10}{10} - k = 1 - k$$

$$u_3 = \frac{10}{u_2} - k = \frac{10}{1-k} - k$$

$$\Sigma = 0 \Rightarrow 10 + (1-k) + \left(\frac{10}{1-k} - k\right) = 0$$

$$\Rightarrow 11 - 2k + \frac{10}{1-k} = 0 \Rightarrow (11-2k)(1-k) + 10 = 0$$

$$\Rightarrow 2k^2 - 13k + 11 + 10 = 0 \Rightarrow 2k^2 - 13k + 21 = 0 \Rightarrow k = 3 \text{ or } k = \frac{7}{2}$$

19(a))

$$u_1 = 1 \qquad u_2 = -7.7 \qquad u_3 = -10.31 \qquad u_4 = -11.093$$

...

$$u_{10} = -11.4283\ldots$$

This sequence is converging as the absolute differences between each successive term are decreasing.

19(b))

$$u_1 = 2 \qquad u_2 = -0.8 \qquad u_3 = -3.88 \qquad u_4 = -7.268$$

...

$$u_{10} = -36.0225\ldots$$

This sequence is diverging as the absolute differences between each successive term are not decreasing.

19(c))

$$u_1 = -3 \qquad u_2 = -5.5 \qquad u_3 = -11.75 \qquad u_4 = -27.375$$

...

$$u_{10} = -6359.1621\ldots$$

This sequence is diverging as the absolute differences between each successive term are not decreasing.

19(d))

$$u_1 = 4 \qquad u_2 = -4.2 \qquad u_3 = 10.56 \qquad u_4 = -16.008$$

...

$$u_{10} = -579.8379\ldots$$

This sequence is diverging as the absolute differences between each successive term are not decreasing.

19(e))

$$u_1 = 10 \qquad u_2 = 91 \qquad u_3 = 18.1 \qquad u_4 = 83.71$$

...

$$u_{10} = 69.1479\ldots$$

This sequence is converging as the absolute differences between each successive term are decreasing.

20(a))

$$u_1 = 3 + 0.6 \times 1 = 3.6 \qquad u_2 = 3 + 0.6 \times 2 = 4.2$$
$$u_3 = 3 + 0.6 \times 3 = 4.8 \qquad u_4 = 3 + 0.6 \times 4 = 5.4$$

...

$$u_{10} = 3 + 0.6 \times 10 = 9$$

This sequence is diverging as the absolute differences between each successive term are not decreasing.

20(b))

$$u_1 = 2^1 = 2 \qquad u_2 = 2^2 = 4 \qquad u_3 = 2^3 = 8 \qquad u_4 = 2^4 = 16$$

...

$$u_{10} = 2^{10} = 1024$$

This sequence is diverging as the absolute differences between each successive term are not decreasing.

20(c))

$$u_1 = \left(\frac{1}{3}\right)^1 = \frac{1}{3} \qquad u_2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$
$$u_3 = \left(\frac{1}{3}\right)^3 = \frac{1}{27} \qquad u_4 = \left(\frac{1}{3}\right)^4 = \frac{1}{81}$$

...

$$u_{10} = \left(\frac{1}{3}\right)^{10} = \frac{1}{59049}$$

This sequence is converging as the absolute differences between each successive term are decreasing.

20(d))

$$u_1 = \frac{10}{1+1} = 5 \qquad u_2 = \frac{10}{2+1} = \frac{10}{3} = 3.3333\ldots$$
$$u_3 = \frac{10}{3+1} = \frac{5}{2} = 2.5 \qquad u_4 = \frac{10}{4+1} = 2$$

...

$$u_{10} = \frac{10}{10+1} = \frac{10}{11} = 0.9090\ldots$$

This sequence is converging as the absolute differences between each successive term are decreasing.

21(a))

$$L = \frac{L}{3} + 50 \Rightarrow 3L = L + 150 \Rightarrow 2L = 150 \Rightarrow L = 75$$

21(b))

$$L = 0.85L - 6 \Rightarrow 0.15L = -6 \Rightarrow L = -\frac{6}{0.15} = -40$$

21(c))

$$L = \frac{L + 15}{30} \Rightarrow 30L = L + 15 \Rightarrow 29L = 15 \Rightarrow L = \frac{15}{29}$$

21(d))

$$L = \frac{0.2L - 100}{25} \Rightarrow 25L = 0.2L - 100$$
$$\Rightarrow 24.8L = -100 \Rightarrow L = -\frac{100}{24.8} = -\frac{125}{31}$$

22(a))

One quarter of the painkiller already in Arnold's system is broken down over the four-hour period.

22(b))

Initial dose = 400 mg.

After 4 hours = $0.25 \times 400 + 400 = 500$ mg.

After 8 hours = $0.25 \times 500 + 400 = 525$ mg.

After 12 hours = $0.25 \times 525 + 400 = 531.25$ mg

= 531 mg to the nearest mg.

22(c))

$$L = 0.25L + 400 \Rightarrow 0.75L = 400$$
$$\Rightarrow L = \frac{400}{0.75} = \frac{1600}{3} = 533.3333\ldots \text{ mg}$$

23(a))

$$u_1 = \frac{9u_1}{8} + 5 \Rightarrow 8u_1 = 9u_1 + 40 \Rightarrow -u_1 = 40 \Rightarrow u_1 = -40$$

23(b))

$$u_1 = \frac{10 - u_1}{15} \Rightarrow 15u_1 = 10 - u_1 \Rightarrow 16u_1 = 10 \Rightarrow u_1 = \frac{10}{16} = \frac{5}{8}$$

23(c))

$$u_1 = \frac{5}{2} + \frac{21}{u_1} \Rightarrow 2u_1^2 = 5u_1 + 42$$

$$\Rightarrow 2u_1^2 - 5u_1 - 42 = 0 \Rightarrow u_1 = -\frac{7}{2} \text{ or } u_1 = 6$$

23(d))

$$u_1 = \frac{55 - 3u_1 - 8u_1^2}{10u_1} \Rightarrow 10u_1^2 = 55 - 3u_1 - 8u_1^2$$

$$\Rightarrow 18u_1^2 + 3u_1 - 55 = 0 \Rightarrow u_1 = -\frac{11}{6} \text{ or } u_1 = \frac{5}{3}$$

24(a))

$$u_1 = \frac{1}{1+10} = \frac{1}{11} = 0.0909\ldots \qquad u_2 = \frac{1}{2+10} = \frac{1}{12} = 0.0833\ldots$$

$$u_3 = \frac{1}{3+10} = \frac{1}{13} = 0.0769\ldots \qquad u_4 = \frac{1}{4+10} = \frac{1}{14} = 0.0714\ldots$$

...

$$u_{1000} = \frac{1}{1000+10} = \frac{1}{1010} = 0.00099\ldots$$

The sequence is converging on 0.

24(b))

$$u_1 = \frac{1}{1+10} = \frac{1}{11} = 0.0909\ldots \qquad u_2 = \frac{2}{2+10} = \frac{2}{12} = \frac{1}{6} = 0.1666\ldots$$

$$u_3 = \frac{3}{3+10} = \frac{3}{13} = 0.2307\ldots \qquad u_4 = \frac{4}{4+10} = \frac{4}{14} = \frac{2}{7} = 0.2857\ldots$$

...

$$u_{1000} = \frac{1000}{1000+10} = \frac{1000}{1010} = 0.9900\ldots$$

The sequence is converging on 1.

24(c))

$$u_1 = \frac{1+11}{1+10} = \frac{12}{11} = 1.0909\ldots \qquad u_2 = \frac{2+11}{2+10} = \frac{13}{12} = 1.0833\ldots$$

$$u_3 = \frac{3+11}{3+10} = \frac{14}{13} = 1.0769\ldots \qquad u_4 = \frac{4+11}{4+10} = \frac{15}{14} = 1.0714\ldots$$

...

$$u_{1000} = \frac{1000+11}{1000+10} = \frac{1011}{1010} = 1.00099\ldots$$

The sequence is converging on 1.

24(d))

$$u_1 = \frac{2\times 1+11}{1+10} = \frac{13}{11} = 1.1818\ldots \qquad u_2 = \frac{2\times 2+10}{2+10} = \frac{15}{12} = \frac{5}{4} = 1.25$$
$$u_3 = \frac{2\times 3+11}{3+10} = \frac{17}{13} = 1.3076\ldots \qquad u_4 = \frac{2\times 4+11}{4+10} = \frac{19}{14} = 1.3571\ldots$$

$\ldots$

$$u_{1000} = \frac{2\times 1000+11}{1000+10} = \frac{2011}{1010} = 1.9910\ldots$$

The sequence is converging on 2.

24(e))

$$u_1 = \frac{2\times 1+11}{1^2+10} = \frac{13}{11} = 1.1818\ldots \qquad u_2 = \frac{2\times 2+11}{2^2+10} = \frac{15}{14} = 1.0714\ldots$$
$$u_3 = \frac{2\times 3+11}{3^2+10} = \frac{17}{19} = 0.8947\ldots \qquad u_4 = \frac{2\times 4+11}{4^2+10} = \frac{19}{26} = 0.7307\ldots$$

$\ldots$

$$u_{1000} = \frac{2\times 1000+11}{1000^2+10} = \frac{12011}{1000010} = 0.0120\ldots$$

The sequence is converging on 0.

25)

$$u_1 = 1 \qquad u_2 = 1-u_1 = 1-1 = 0$$
$$u_3 = 1-u_2 = 1-0 = 1 \qquad u_4 = 1-u_3 = 1-1 = 0$$

$\ldots$

$$u_{155} = 1$$

(odd n gives 1, even n gives 0)

26)

$$u_1 = 1 \qquad u_2 = 8-u_1 = 8-1 = 7$$
$$u_3 = 8-u_2 = 8-7 = 1 \qquad u_4 = 8-u_3 = 8-1 = 7$$

$\ldots$

$$u_{88} = 7$$

(odd n gives 1, even n gives 7)

27(a))

$$u_1 = 1 \qquad u_2 = -u_1 = -1 \qquad u_3 = -u_2 = 1 \qquad u_4 = -u_3 = -1$$

$\ldots$

$$u_{100} = -1$$

(odd n gives 1, even n gives -1)

27(b))

$$\sum_{n=1}^{100} u_n = 1 + (-1) + 1 + (-1) + \cdots + 1 + (-1)$$
$$= (1 + 1 + \cdots + 1) + ((-1) + (-1) + \cdots + (-1)) = 50 \times 1 + 50 \times (-1) = 0$$

28(a))

$$u_1 = 4 \qquad u_2 = 13 - u_1^2 = 13 - 4^2 = -3$$
$$u_3 = 13 - u_2^2 = 13 - (-3)^2 = 4 \qquad u_4 = 13 - u_3^2 = 13 - 4^2 = -3$$

$\cdots$

$$u_{500} = -3$$

(odd n gives 4, even n gives -3)

28(b))

$$\sum_{n=1}^{500} u_n = 4 + (-3) + 4 + (-3) + \cdots + 4 + (-3)$$
$$= (4 + 4 + \cdots + 4) + ((-3) + (-3) + \cdots + (-3)) = 250 \times 4 + 250 \times (-3) = 250$$

29(a))

$$u_1 = \frac{5}{3} \qquad u_2 = -\frac{1}{u_1} = -\frac{3}{5} \qquad u_3 = -\frac{1}{u_2} = \frac{5}{3} \qquad u_4 = -\frac{1}{u_3} = -\frac{3}{5}$$

$\cdots$

$$u_{99} = \frac{5}{3}$$

(odd n gives $\frac{5}{3}$, even n gives $-\frac{3}{5}$)

29(b))

$$\sum_{n=1}^{99} u_n = \frac{5}{3} + \left(-\frac{3}{5}\right) + \frac{5}{3} + \left(-\frac{3}{5}\right) + \cdots + \frac{5}{3} + \left(-\frac{3}{5}\right) + \frac{5}{3}$$
$$= \left(\frac{5}{3} + \frac{5}{3} + \cdots + \frac{5}{3} + \frac{5}{3}\right) + \left(\left(-\frac{3}{5}\right) + \left(-\frac{3}{5}\right) + \cdots + \left(-\frac{3}{5}\right)\right)$$
$$= 50 \times \frac{5}{3} + 49 \times \left(-\frac{3}{5}\right) = \frac{250}{3} - \frac{147}{5} = \frac{809}{15}$$

30(a))

$$u_1 = 3 \qquad u_2 = \frac{u_1 - 1}{k u_1} = \frac{3 - 1}{k \times 3} = \frac{2}{3k}$$
$$u_3 = \frac{u_2 - 1}{k u_2} = \frac{\frac{2}{3k} - 1}{k \times \frac{2}{3k}} = \frac{\frac{2}{3k} - 1}{2/3} = \frac{1}{k} - \frac{3}{2}$$

$$u_4 = \frac{u_3 - 1}{ku_3} = \frac{\frac{1}{k} - \frac{3}{2} - 1}{k\left(\frac{1}{k} - \frac{3}{2}\right)} = \frac{\frac{1}{k} - \frac{5}{2}}{1 - \frac{3}{2}k} = \frac{2 - 5k}{2k - 3k^2}$$

If the sequence is periodic with period 3 then $u_1 = u_4$.

$$\frac{2 - 5k}{2k - 3k^2} = 3 \Rightarrow 2 - 5k = 6k - 9k^2 \Rightarrow 9k^2 - 11k + 2 = 0 \Rightarrow k = \frac{2}{9} \text{ or } k = 1$$

If $k = \frac{2}{9}$ then $u_1 = 3$, $u_2 = 3$, $u_3 = 3$, $u_4 = 3$, ... which is a constant sequence and so is not a periodic sequence with period 3.

If $k = 1$ then $u_1 = 3$, $u_2 = \frac{2}{3}$, $u_3 = -\frac{1}{2}$, $u_4 = 3$, ... which is a periodic sequence with period 3, as required. So $k = 1$.

30(b))

$$u_1 = 3 \qquad u_2 = \frac{2}{3} \qquad u_3 = -\frac{1}{2}$$

...

$$\frac{100}{3} = 33 \text{ r } 1$$

$$\sum_{n=1}^{100} u_n = \left(3 + \frac{2}{3} + \left(-\frac{1}{2}\right)\right) + \left(3 + \frac{2}{3} + \left(-\frac{1}{2}\right)\right) + \cdots + \left(3 + \frac{2}{3} + \left(-\frac{1}{2}\right)\right) + 3$$

$$= 33\left(3 + \frac{2}{3} + \left(-\frac{1}{2}\right)\right) + 3 = \frac{215}{2}$$

31)

$$u_1 = 4 \qquad u_2 = -6 \qquad u_3 = 8$$

$$u_4 = u_1 = 4 \qquad u_5 = u_2 = -6 \qquad u_6 = u_3 = 8$$

...

$$\frac{500}{3} = 166 \text{ r } 2$$

$$\sum_{n=1}^{500} u_n = (4 + (-6) + 8) + (4 + (-6) + 8) + \cdots + (4 + (-6) + 8) + 4 + (-6)$$

$$= 166(4 + (-6) + 8) + 4 + (-6) = 994$$

32)

$$u_1 = 1 \qquad u_2 = 4 \qquad u_3 = -7 \qquad u_4 = -2$$

$$u_5 = u_1 = 1 \qquad u_6 = u_2 = 4 \qquad u_7 = u_3 = -7 \qquad u_8 = u_4 = -2$$

...

$$\frac{499}{4} = 124 \text{ r } 3$$

$$\sum_{n=1}^{499} u_n = (1 + 4 + (-7) + (-2)) + (1 + 4 + (-7) + (-2)) + \cdots + (1 + 4 + (-7) + (-2)) + 1 + 4 + (-7)$$

$$= 124(1 + 4 + (-7) + (-2)) + 1 + 4 + (-7) = -498$$

33)

$$u_1 = 0 \qquad u_2 = 1 \qquad u_3 = 5$$

$$u_4 = u_1 + 2 = 0 + 2 = 2 \qquad u_5 = u_2 + 2 = 1 + 2 = 3 \qquad u_6 = u_3 + 2 = 5 + 2 = 7$$

$$u_7 = 4,\ u_{10} = 6,\ u_{13} = 8,\ u_{16} = 10$$

...

So only the terms u_1, u_4, u_7, u_{10}, u_{13} etc will be even, and the rest will be odd, as we started with an even number and are adding on 2 each time.

So let's focus on these terms:

$$u_1 = 0,\ u_4 = 2,\ u_7 = 4,\ u_{10} = 6,\ u_{13} = 8,\ u_{16} = 10$$

Let's create a new sequence, p_m:

$$p_1 = 0,\ p_2 = 2,\ p_3 = 4,\ p_4 = 6,\ p_5 = 8,\ p_6 = 10$$

So that $p_m = u_{3m-2}$.

This means that $p_6 = u_{3\times 6-2} = u_{16}$.

$p_m = 2m - 2$ is the mth term of this sequence, so $2m - 2 = 200 \Rightarrow m = 101$.

If $p_{101} = 200$ then $200 = u_{3\times 101-2} = u_{301}$.

So the 301st term in the original sequence is 200.

$$\therefore k = 301$$

34)

$$u_1 = 3 + (-1)^1 = 2 \qquad u_2 = 3 + (-1)^2 = 4$$

$$u_3 = 3 + (-1)^3 = 2 \qquad u_4 = 3 + (-1)^4 = 4$$

...

This sequence has period 2.

35)

$$u_1 = \sin\left(\frac{3\pi \times 1}{2}\right) = -1 \qquad u_2 = \sin\left(\frac{3\pi \times 2}{2}\right) = 0$$

$$u_3 = \sin\left(\frac{3\pi \times 3}{2}\right) = 1 \qquad u_4 = \sin\left(\frac{3\pi \times 4}{2}\right) = 0$$

$$u_5 = \sin\left(\frac{3\pi \times 5}{2}\right) = -1$$

...

This sequence has period 4.

36)

$$u_1 = \cos\left(\frac{2\pi \times 1}{3}\right) = -\frac{1}{2} \qquad u_2 = \cos\left(\frac{2\pi \times 2}{3}\right) = -\frac{1}{2}$$
$$u_3 = \cos\left(\frac{2\pi \times 3}{3}\right) = 1 \qquad u_4 = \cos\left(\frac{2\pi \times 4}{3}\right) = -\frac{1}{2}$$
$$u_5 = \cos\left(\frac{2\pi \times 5}{3}\right) = -\frac{1}{2} \qquad u_6 = \cos\left(\frac{2\pi \times 6}{3}\right) = 1$$

...

This sequence has period 3.

37)

$$u_1 = \sin\left(\frac{2\pi \times 1}{3}\right) - \cos\left(\frac{5\pi \times 1}{3}\right) = \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$$
$$u_2 = \sin\left(\frac{2\pi \times 2}{3}\right) - \cos\left(\frac{5\pi \times 2}{3}\right) = -\frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{1-\sqrt{3}}{2}$$
$$u_3 = \sin\left(\frac{2\pi \times 3}{3}\right) - \cos\left(\frac{5\pi \times 3}{3}\right) = 0 + 1 = 1$$
$$u_4 = \sin\left(\frac{2\pi \times 4}{3}\right) - \cos\left(\frac{5\pi \times 4}{3}\right) = \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3}+1}{2}$$
$$u_5 = \sin\left(\frac{2\pi \times 5}{3}\right) - \cos\left(\frac{5\pi \times 5}{3}\right) = -\frac{\sqrt{3}}{2} - \frac{1}{2} = -\frac{\sqrt{3}+1}{2}$$
$$u_6 = \sin\left(\frac{2\pi \times 6}{3}\right) - \cos\left(\frac{5\pi \times 6}{3}\right) = 0 - 1 = -1$$
$$u_7 = \sin\left(\frac{2\pi \times 7}{3}\right) - \cos\left(\frac{5\pi \times 7}{3}\right) = \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$$

...

This sequence has period 6.

38)

$$u_1 = \cos\left(\frac{\pi \times 1}{4}\right) - \sin\left(\frac{5\pi \times 1}{2}\right) = \frac{\sqrt{2}}{2} - 1 = \frac{\sqrt{2}-2}{2}$$
$$u_2 = \cos\left(\frac{\pi \times 2}{4}\right) - \sin\left(\frac{5\pi \times 2}{2}\right) = 0 - 0 = 0$$
$$u_3 = \cos\left(\frac{\pi \times 3}{4}\right) - \sin\left(\frac{5\pi \times 3}{2}\right) = -\frac{\sqrt{2}}{2} + 1 = \frac{2-\sqrt{2}}{2}$$
$$u_4 = \cos\left(\frac{\pi \times 4}{4}\right) - \sin\left(\frac{5\pi \times 4}{2}\right) = -1 - 0 = -1$$
$$u_5 = \cos\left(\frac{\pi \times 5}{4}\right) - \sin\left(\frac{5\pi \times 5}{2}\right) = -\frac{\sqrt{2}}{2} - 1 = -\frac{2+\sqrt{2}}{2}$$
$$u_6 = \cos\left(\frac{\pi \times 6}{4}\right) - \sin\left(\frac{5\pi \times 6}{2}\right) = 0 - 0 = 0$$
$$u_7 = \cos\left(\frac{\pi \times 7}{4}\right) - \sin\left(\frac{5\pi \times 7}{2}\right) = \frac{\sqrt{2}}{2} + 1 = \frac{2+\sqrt{2}}{2}$$

$$u_8 = \cos\left(\frac{\pi \times 8}{4}\right) - \sin\left(\frac{5\pi \times 8}{2}\right) = 1 - 0 = 1$$
$$u_9 = \cos\left(\frac{\pi \times 9}{4}\right) - \sin\left(\frac{5\pi \times 9}{2}\right) = \frac{\sqrt{2}}{2} - 1 = \frac{\sqrt{2} - 2}{2}$$

...

This sequence has period 8.

39)

$$u_1 = (-1)^1 + \sin\left(\frac{\pi \times 1}{2}\right) = 0 \qquad u_2 = (-1)^2 + \sin\left(\frac{\pi \times 2}{2}\right) = 1$$
$$u_3 = (-1)^3 + \sin\left(\frac{\pi \times 3}{2}\right) = -2 \qquad u_4 = (-1)^4 + \sin\left(\frac{\pi \times 4}{2}\right) = 1$$
$$u_5 = (-1)^5 + \sin\left(\frac{\pi \times 5}{2}\right) = 0$$

...

This sequence has period 4.

40)

$$u_1 = 3(-1)^1 + \cos\left(\frac{\pi \times 1}{3}\right) = -\frac{5}{2} \qquad u_2 = 3(-1)^2 + \cos\left(\frac{\pi \times 2}{3}\right) = \frac{5}{2}$$
$$u_3 = 3(-1)^3 + \cos\left(\frac{\pi \times 3}{3}\right) = -4 \qquad u_4 = 3(-1)^4 + \cos\left(\frac{\pi \times 4}{3}\right) = \frac{5}{2}$$
$$u_5 = 3(-1)^5 + \cos\left(\frac{\pi \times 5}{3}\right) = -\frac{5}{2}$$

...

This sequence has period 4.

41)

$$u_1 = 2(-1)^1 - \frac{1}{\cos(\pi \times 1/3)} = -4 \qquad u_2 = 2(-1)^2 - \frac{1}{\cos(\pi \times 2/3)} = 4$$
$$u_3 = 2(-1)^3 - \frac{1}{\cos(\pi \times 3/3)} = -1 \qquad u_4 = 2(-1)^4 - \frac{1}{\cos(\pi \times 4/3)} = 4$$
$$u_5 = 2(-1)^5 - \frac{1}{\cos(\pi \times 5/3)} = -4 \qquad u_6 = 2(-1)^6 - \frac{1}{\cos(\pi \times 6/3)} = 1$$
$$u_7 = 2(-1)^7 - \frac{1}{\cos(\pi \times 7/3)} = -4$$

...

This sequence has period 6.

42)

$$u_1 = \frac{3 \times 1}{2 + 1^2} = \frac{3}{3} = 1 \qquad u_2 = \frac{3 \times 2}{2 + 2^2} = \frac{6}{6} = 1 \qquad u_3 = \frac{3 \times 3}{2 + 3^2} = \frac{9}{11}$$

$u_3 = \frac{9}{11} \not> u_2 = 1$ so this is not an increasing sequence.

43)

$$u_1 = \frac{1+1}{1^3+1-1} = \frac{2}{1} = 2 \qquad u_2 = \frac{2+1}{2^3+2-1} = \frac{1}{3}$$

$u_2 = \frac{1}{3} \not> u_1 = 2$ so this is not an increasing sequence.

44)

$$u_1 = 1 - \frac{1}{1+1} = \frac{1}{2} \qquad u_2 = 2 - \frac{1}{2+1} = \frac{5}{3}$$

$u_2 = \frac{5}{3} \not< u_1 = \frac{1}{2}$ so this is not a decreasing sequence.

45)

$$u_{100} = 100^2 - 100 \times 100 = 0 \qquad u_{101} = 101^2 - 100 \times 101 = 101$$

$u_{101} = 101 \not< u_{100} = 0$ so this is not a decreasing sequence.

46)

u_n is a decreasing sequence if $u_{n+1} - u_n < 0$ for all integers $n \geq 1$.

$$3^{-(n+1)} - 3^{-n} = \frac{1}{3^{n+1}} - \frac{1}{3^n} = \frac{1}{3^{n+1}} - \frac{3}{3^{n+1}} = -\frac{2}{3^{n+1}}$$

$3^{n+1} > 0$ for all integers $n \geq 1$.

$$\therefore -\frac{2}{3^{n+1}} < 0 \text{ for all integers } n \geq 1$$

Hence u_n is a decreasing sequence.

47)

u_n is a decreasing sequence if $u_{n+1} - u_n < 0$ for all integers $n \geq 1$.

$$\frac{1}{(n+1)^2} - \frac{1}{n^2} = \frac{n^2-(n+1)^2}{(n+1)^2n^2} = \frac{n^2-(n^2+2n+1)}{(n+1)^2n^2} = -\frac{2n+1}{(n+1)^2n^2}$$

$2n+1 > 0$, $(n+1)^2 > 0$ and $n^2 > 0$ for all integers $n \geq 1$.

$$\therefore -\frac{2n+1}{(n+1)^2n^2} < 0 \text{ for all integers } n \geq 1$$

Hence u_n is a decreasing sequence.

48)

u_n is an increasing sequence if $u_{n+1} - u_n > 0$ for all integers $n \geq 1$.

$$\frac{(n+1)+1}{(n+1)+2} - \frac{n+1}{n+2} = \frac{n+2}{n+3} - \frac{n+1}{n+2} = \frac{(n+2)^2-(n+1)(n+3)}{(n+3)(n+2)}$$

$$= \frac{n^2+4n+4-(n^2+4n+3)}{(n+3)(n+2)} = \frac{1}{(n+3)(n+2)}$$

$n+3 > 0$ and $n+2 > 0$ for all integers $n \geq 1$.

$$\therefore \frac{1}{(n+3)(n+2)} > 0 \text{ for all integers } n \geq 1$$

Hence u_n is an increasing sequence.

49)

u_n is a decreasing sequence if $u_{n+1} - u_n < 0$ for all integers $n \geq 1$.

$$\frac{(n+1)+3}{(n+1)+2} - \frac{n+3}{n+2} = \frac{n+4}{n+3} - \frac{n+3}{n+2} = \frac{(n+4)(n+2)-(n+3)^2}{(n+3)(n+2)}$$

$$= \frac{(n^2+6n+8)-(n^2+6n+9)}{(n+3)(n+2)} = -\frac{1}{(n+3)(n+2)}$$

$n+3 > 0$ and $n+2 > 0$ for all integers $n \geq 1$.

$$\therefore -\frac{1}{(n+3)(n+2)} < 0 \text{ for all integers } n \geq 1$$

Hence u_n is a decreasing sequence.

50)

u_n is a decreasing sequence if $u_{n+1} - u_n < 0$ for all integers $n \geq 1$.

$$\frac{5}{(n+1)+5} - \frac{5}{n+5} = \frac{5}{n+6} - \frac{5}{n+5} = \frac{5(n+5)-5(n+6)}{(n+6)(n+5)} = -\frac{5}{(n+6)(n+5)}$$

$n+6 > 0$ and $n+5 > 0$ for all integers $n \geq 1$.

$$\therefore -\frac{5}{(n+6)(n+5)} < 0 \text{ for all integers } n \geq 1$$

Hence u_n is a decreasing sequence.

51)

u_n is a decreasing sequence if $u_{n+1} - u_n < 0$ for all integers $n \geq 1$.

$$\frac{2-(n+1)}{n+1} - \frac{2-n}{n} = \frac{1-n}{n+1} - \frac{2-n}{n} = \frac{(1-n)n-(n+1)(2-n)}{(n+1)n}$$

$$= \frac{n-n^2-(-n^2+n+2)}{(n+1)n} = -\frac{2}{(n+1)n}$$

$n+1 > 0$ and $n > 0$ for all integers $n \geq 1$.

$$\therefore -\frac{2}{(n+1)n} < 0 \text{ for all integers } n \geq 1$$

Hence u_n is a decreasing sequence.

52)

u_n is an increasing sequence if $u_{n+1} - u_n > 0$ for all integers $n \geq 1$.

$$\left((2(n+1)+1)^2 - ((n+1)-1)^2\right) - \left((2n+1)^2 - (n-1)^2\right)$$

$$= \left((2n+3)^2 - n^2\right) - \left((2n+1)^2 - (n-1)^2\right)$$

$$= (4n^2+12n+9-n^2) - ((4n^2+4n+1)-(n^2-2n+1))$$

$$= (3n^2+12n+9) - (3n^2+6n) = 6n+9$$

$6n+9 > 0$ for all integers $n \geq 1$.

Hence u_n is an increasing sequence.

53)

u_n is an increasing sequence if $u_{n+1} - u_n > 0$ for all integers $n \geq 1$.

$$\left(\sqrt{(n+1)+1} + \sqrt{n+1}\right) - \left(\sqrt{n+1} + \sqrt{n}\right)$$

$$= \sqrt{n+2} + \sqrt{n+1} - \sqrt{n+1} - \sqrt{n} = \sqrt{n+2} - \sqrt{n}$$

$n + 2 > n$ so $\sqrt{n+2} > \sqrt{n}$.

So $\sqrt{n+2} - \sqrt{n} > 0$ for all integers $n \geq 1$.

Hence u_n is an increasing sequence.